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System operation:

Information about successful start of motor is got from auxiliary contact of the contactor K1 (I0.0 (motor running = signal true)).

At the first start, a monitoring time of ten seconds is started. If three unsuccessful start attempts are detected during the monitoring time, the system generates an alarm "ALARM" by setting output Q 4.0 on. If fewer than three unsuccessful start attempts are detected during the monitoring time, no alarm is generated. A successful start here means that the motor control is on, and feedback information is available from the start. The control and feedback must remain on for one second for the start to be considered successful. If the motor is started but no feedback is received for more than one second, the start attempt is considered unsuccessful.

All start attempts are counted and the number of them "COUNT" is always visible in the output byte QB 5. The set alarm (Q 4.0) is reset "RESET" via input I 0.7 with a rising edge of the switch signal. In this application, a two-position toggle switch is available for resetting the alarm (bistable). To avoid a situation where a reset switch that has been forgotten to be in the ON position would prevent the alarm from being generated, the rising edge of the switch signal (impulse) is used for resetting.

Task: Create a program that implements the function as told. The motor is controlled by the start switch I0.1, the stop by the "run permission" switch I0.2. Contactor K1 is connected to output address Q4.1 When run permission is removed (the signal is false), the motor must stop. Similarly, an active alarm must control the motor to stop.

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